Table 1: Histogram-production workflow in execution order.

Step	Operation	Action	Selection/correction status
1	Input skim chunks	Read validated Events trees; MC normalization denominators come from dataset-wide skim reports/manifest metadata	No new object calibration; consumes corrected and shifted skim branches
2	Optional additional cut	If –additional-cuts is supplied, apply it directly with RDataFrame.Filter	User-defined event filter, before derived observables and weights
3	Weights and observables	Build central/shifted weights, trigger SF, PDF and QCD-scale weights; define selected-jet, VBF, dimuon, soft-activity and requested DNN observables	Uses corrections stored by the skim
4	Shifted jet observables	For requested JER/JES suffixes, rebuild selected-jet and VBF observables; fail if required skim columns are absent	Shape-systematic preparation
5	Reco/gen jet matching	For eligible MC, define jet matching and optional DY/EWK exclusive jet-component categories	MC only; optional component workflow
6	Histogram selections	Define/redefine every masses_regions, muons_selection, jets_selection and categories expression from selections.yaml, including requested shape suffixes	Definitions first; no rejection yet
7	DY normalization/reweights	Apply amcatnlo normalization and optional DY pT(ll) and Njets reweights to all target weight columns	MC DY only; weights, not filters
8	Mass-region filter	Filter on each requested stored mass-region column (default: mass_inclusive, Z_sideband, Signal_Fit)	First histogram-level event selection
9	Sideband DNN	For DNN_NNOutput in Z/H sidebands, remap m_mumu into the training window and reevaluate the dedicated DNN payload	Only requested DNN sideband histograms
10	Category filter	Within each mass region, filter each requested stored category (default: baseline, ggF, VBF)	Second histogram-level event selection
11	Systematic routing	Choose shape-shifted variable/selection columns or weight-only variations; data are forced to Central	JER/JES/Muon scale/resolution shapes; PU/muon/PDF/QCD weights
12	Histogram booking	Book configured 1D/2D models with the selected variable and weight; missing shifted variables produce empty templates	Output organized by mass region and category
13	Chunk merge/postprocessing	Merge temporary ROOT outputs; build QCD scale envelopes, PDF treatment and optional split component files	Final histogram products

Table 2: All histogram-level selection definitions. Rows are merged when identical across eras.

YAML section	Name	Expression	Store	Eras	Histogram use
masses_regions	Z_sideband	m_mumu{mu_suff} > 70 && m_mumu{mu_suff} < 110	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
masses_regions	Signal_Fit	m_mumu{mu_suff} > 115 && m_mumu{mu_suff} < 135	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
masses_regions	H_sideband	((m_mumu{mu_suff} > 110 && m_mumu{mu_suff} < 115) || (m_mumu{mu_suff} > 135 && m_mumu{mu_suff} < 150))	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
masses_regions	Signal_ext	m_mumu{mu_suff} > 110 && m_mumu{mu_suff} < 150	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
masses_regions	mass_inclusive	m_mumu{mu_suff} > 60 && m_mumu{mu_suff} < 150	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
muons_selection	muon_pre_sel_check	mu1_pt{mu_suff} > 15 && abs(mu1_eta) < 2.4 && mu2_pt{mu_suff} > 15 && abs(mu2_eta) < 2.4	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	muons_OS	mu1_charge{mu_suff} * mu2_charge{mu_suff} < 0	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	muons_OS_and_presel	muon_pre_sel_check{mu_suff} && muons_OS{mu_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	muons_trigger	(Event_HasTriggerMatching_singleMu{mu_suff} && HLT_IsoMu24 && ((mu1_pt{mu_suff} > 26 && mu1_HasTriggerMatching_singleMu{mu_suff}) || (mu2_pt{mu_suff} > 26 && mu2_HasTriggerMatching_singleMu{mu_suff})))	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	muons_OS_presel_trg	muons_OS_and_presel{mu_suff} && muons_trigger{mu_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	mu1_pt_l30	mu1_pt{mu_suff} < 30	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	mu2_pt_l30	mu2_pt{mu_suff} < 30	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	mu1_pt_g20	mu1_pt{mu_suff} > 20	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	mu2_pt_g20	mu2_pt{mu_suff} > 20	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	mu_pt_sel	mu1_pt_g20{mu_suff} && mu2_pt_g20{mu_suff}	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	mu1_ML	mu1_pfIsoId{mu_suff} >= 2 && mu1_mediumId{mu_suff}	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	mu1_LL	mu1_pfIsoId{mu_suff} >= 2 && mu1_looseId{mu_suff}	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	mu1_TT	mu1_pfIsoId{mu_suff} >= 4 && mu1_tightId{mu_suff}	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	mu2_ML	mu2_pfIsoId{mu_suff} >= 2 && mu2_mediumId{mu_suff}	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	mu2_LL	mu2_pfIsoId{mu_suff} >= 2 && mu2_looseId{mu_suff}	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	mu2_TT	mu2_pfIsoId{mu_suff} >= 4 && mu2_tightId{mu_suff}	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	bothMuons_ML	mu1_ML{mu_suff} && mu2_ML{mu_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency

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Table 2 – continued

YAML section	Name	Expression	Store	Eras	Histogram use
muons_selection	bothMuons_LL	mu1_LL{mu_suff} && mu2_LL{mu_suff}	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	chi2_sel	mu1_bsConstrainedChi2{mu_suff} < 30 && mu2_bsConstrainedChi2{mu_suff} < 30	yes	2022, 2022EE, 2023, 2023BPix, 2025, 2026	definition/dependency
muons_selection	pt_lessthan30_and_TT	(((mu1_pt_l30{mu_suff} && mu1_TT{mu_suff}) || !(mu1_pt_l30{mu_suff})) && ((mu2_pt_l30{mu_suff} && mu2_TT{mu_suff}) || !(mu2_pt_l30{mu_suff})))	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
jets_selection	has_good_jets	N_SelectedJets{jet_suff} >= 0	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
jets_selection	jets_btag_clean_sel	SelectedJetTagSel{jet_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
categories	base_sel	muons_OS_presel_trg{mu_suff} && mu_pt_sel{mu_suff} && SelectedJetTagSel{jet_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	baseline	base_sel{tot_suff} && mu1_ML{mu_suff} && mu2_ML{mu_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2026	filter candidate
categories	baseline_chi2	baseline{tot_suff} && chi2_sel{mu_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2025, 2026	filter candidate
categories	baseline_lowPtTT	baseline{tot_suff} && pt_lessthan30_and_TT{mu_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2026	filter candidate
categories	baseline_chi2_lowPtTT	baseline{tot_suff} && pt_lessthan30_and_TT{mu_suff} && chi2_sel{mu_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2026	filter candidate
categories	VBF_def	HasVBF{jet_suff} && SelectedJet_pt{jet_suff}[VBFJetIdx_1{jet_suff}] >= 35 && SelectedJet_pt{jet_suff}[VBFJetIdx_2{jet_suff}] >= 25	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	VBF	baseline{tot_suff} && VBF_def{tot_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	ggF	baseline{tot_suff} && !(VBF_def{tot_suff})	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	ggF_0J	ggF{tot_suff} && N_SelectedJets{jet_suff} == 0	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	ggF_1J	ggF{tot_suff} && N_SelectedJets{jet_suff} == 1	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	ggF_ge2J	ggF{tot_suff} && N_SelectedJets{jet_suff} >= 2	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	VBF_ge2J	VBF{tot_suff} && N_SelectedJets{jet_suff} >= 2	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	VBF_lowPtTT	baseline_lowPtTT{tot_suff} && VBF_def{tot_suff}	yes	2022	filter candidate
categories	ggF_lowPtTT	baseline_lowPtTT{tot_suff} && !(VBF_def{tot_suff})	yes	2022	filter candidate
categories	ggF_Z_0J	ggF_0J{tot_suff}	yes	2022EE, 2023, 2023BPix, 2025, 2026	filter candidate
categories	ggF_Z_1J	ggF_1J{tot_suff}	yes	2022EE, 2023, 2023BPix, 2025, 2026	filter candidate
categories	ggF_Z_ge2J	ggF_ge2J{tot_suff}	yes	2022EE, 2023, 2023BPix, 2025, 2026	filter candidate
categories	VBF_Z_ge2J	VBF_ge2J{tot_suff}	yes	2022EE, 2023, 2023BPix, 2025, 2026	filter candidate
categories	baseline	base_sel{tot_suff} && mu1_ML{mu_suff} && mu2_ML{mu_suff} && pt_lessthan30_and_TT{mu_suff}	yes	2024, 2025	filter candidate

Table 3: Shape and weight systematics configured for histogram production.

Type	Systematic	Muon suffix	Jet suffix / weight expression	Weight column	Eras
shape	Central			weight___Central	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
shape	JER		_JER{scale}	weight___Central	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
shape	JES_Total		_JESTotal{scale}	weight___Central	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
shape	MuonScale	_FSR_scale_{scale}		weight___Central	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
shape	MuonRes	_FSR_res_{scale}		weight___Central	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
weight	Central		genWeight * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrgSF_singleMu_IsoMu24Central	weight___Central	2022, 2022EE, 2023, 2023BPix

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Table 3 – continued

Type	Systematic	Muon suffix	Jet suffix / weight expression	Weight column	Eras
weight	QCDScale__{scale}		derived/envelope	weight___QCDScale__{scale}	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
weight	PDF__{scale}		weight__pdf__{scale}	weight___PDF__{scale}	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
weight	PU__{scale}		genWeight * weight__lumi * weight__xs * weight__pu__{scale} * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24Central	weight___PU__{scale}	2022, 2022EE, 2023, 2023BPix
weight	MuonID__{scale}		genWeight * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__{scale} * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__{scale} * weight__TrgSF__singleMu__IsoMu24Central	weight___MuonID__{scale}	2022, 2022EE, 2023, 2023BPix
weight	MuonIso__{scale}		genWeight * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__{scale} * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__{scale} * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24Central	weight___MuonIso__{scale}	2022, 2022EE, 2023, 2023BPix
weight	singleMuTrigger__{scale}		genWeight * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24{scale}	weight___singleMuTrigger__{scale}	2022, 2022EE, 2023, 2023BPix
weight	Central		genWeight * weight__lumi * weight__xs * weight__pu * ((mu1__pt < 30) ? weight__mu1__TightPFIso__TightID__Central : weight__mu1__LoosePFIso__MediumID__Central) * ((mu1__pt < 30) ? weight__mu1__TightID__Trk__Central : weight__mu1__MediumID__Trk__Central) * ((mu2__pt < 30) ? weight__mu2__TightPFIso__TightID__Central : weight__mu2__LoosePFIso__MediumID__Central) * ((mu2__pt < 30) ? weight__mu2__TightID__Trk__Central : weight__mu2__MediumID__Trk__Central) * weight__TrgSF__singleMu__IsoMu24Central	weight___Central	2024, 2025
weight	PU__{scale}		genWeight * weight__lumi * weight__xs * weight__pu__{scale} * ((mu1__pt < 30) ? weight__mu1__TightPFIso__TightID__Central : weight__mu1__LoosePFIso__MediumID__Central) * ((mu1__pt < 30) ? weight__mu1__TightID__Trk__Central : weight__mu1__MediumID__Trk__Central) * ((mu2__pt < 30) ? weight__mu2__TightPFIso__TightID__Central : weight__mu2__LoosePFIso__MediumID__Central) * ((mu2__pt < 30) ? weight__mu2__TightID__Trk__Central : weight__mu2__MediumID__Trk__Central) * weight__TrgSF__singleMu__IsoMu24Central	weight___PU__{scale}	2024, 2025
weight	MuonID__{scale}		genWeight * weight__lumi * weight__xs * weight__pu * ((mu1__pt < 30) ? weight__mu1__TightPFIso__TightID__Central : weight__mu1__LoosePFIso__MediumID__Central) * ((mu1__pt < 30) ? weight__mu1__TightID__Trk__{scale} : weight__mu1__MediumID__Trk__{scale}) * ((mu2__pt < 30) ? weight__mu2__TightPFIso__TightID__Central : weight__mu2__LoosePFIso__MediumID__Central) * ((mu2__pt < 30) ? weight__mu2__TightID__Trk__{scale} : weight__mu2__MediumID__Trk__{scale}) * weight__TrgSF__singleMu__IsoMu24Central	weight___MuonID__{scale}	2024, 2025
weight	MuonIso__{scale}		genWeight * weight__lumi * weight__xs * weight__pu * ((mu1__pt < 30) ? weight__mu1__TightPFIso__TightID__{scale} : weight__mu1__LoosePFIso__MediumID__{scale}) * ((mu1__pt < 30) ? weight__mu1__TightID__Trk__Central : weight__mu1__MediumID__Trk__Central) * ((mu2__pt < 30) ? weight__mu2__TightPFIso__TightID__{scale} : weight__mu2__LoosePFIso__MediumID__{scale}) * ((mu2__pt < 30) ? weight__mu2__TightID__Trk__Central : weight__mu2__MediumID__Trk__Central) * weight__TrgSF__singleMu__IsoMu24Central	weight___MuonIso__{scale}	2024, 2025
weight	singleMuTrigger__{scale}		genWeight * weight__lumi * weight__xs * weight__pu * ((mu1__pt < 30) ? weight__mu1__TightPFIso__TightID__Central : weight__mu1__LoosePFIso__MediumID__Central) * ((mu1__pt < 30) ? weight__mu1__TightID__Trk__Central : weight__mu1__MediumID__Trk__Central) * ((mu2__pt < 30) ? weight__mu2__TightPFIso__TightID__Central : weight__mu2__LoosePFIso__MediumID__Central) * ((mu2__pt < 30) ? weight__mu2__TightID__Trk__Central : weight__mu2__MediumID__Trk__Central) * weight__TrgSF__singleMu__IsoMu24{scale}	weight___singleMuTrigger__{scale}	2024, 2025
weight	Central		weight__gen * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24Central		2026
weight	PU__up		weight__gen * weight__lumi * weight__xs * weight__pu__up * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24Central	weight___PU__up	2026
weight	PU__down		weight__gen * weight__lumi * weight__xs * weight__pu__down * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24Central	weight___PU__down	2026
weight	MuonID__up		weight__gen * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__up * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__up * weight__TrgSF__singleMu__IsoMu24Central	weight___MuonID__up	2026
weight	MuonID__down		weight__gen * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__down * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__down * weight__TrgSF__singleMu__IsoMu24Central	weight___MuonID__down	2026
weight	MuonIso__up		weight__gen * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__up * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__up * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24Central	weight___MuonIso__up	2026

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Table 3 – continued

Type	Systematic	Muon suffix	Jet suffix / weight expression	Weight column	Eras
weight	MuonIso__down		weight__gen * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__down * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__down * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24Central	weight___MuonIso__down	2026
weight	singleMuTrigger__up		weight__gen * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24up	weight___singleMuTrigger__up	2026
weight	singleMuTrigger__down		weight__gen * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24down	weight___singleMuTrigger__down	2026

Table 4: Central event weights used to fill MC histograms.

Era	Central MC weight expression
Run3__2022	genWeight * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24Central
Run3__2022EE	genWeight * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24Central
Run3__2023	genWeight * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24Central
Run3__2023BPix	genWeight * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24Central
Run3__2024	genWeight * weight__lumi * weight__xs * weight__pu * ((mu1__pt < 30) ? weight__mu1__TightPFIso__TightID__Central : weight__mu1__LoosePFIso__MediumID__Central) * ((mu1__pt < 30) ? weight__mu1__TightID__Trk__Central : weight__mu1__MediumID__Trk__Central) * ((mu2__pt < 30) ? weight__mu2__TightPFIso__TightID__Central : weight__mu2__LoosePFIso__MediumID__Central) * ((mu2__pt < 30) ? weight__mu2__TightID__Trk__Central : weight__mu2__MediumID__Trk__Central) * weight__TrgSF__singleMu__IsoMu24Central
Run3__2025	genWeight * weight__lumi * weight__xs * weight__pu * ((mu1__pt < 30) ? weight__mu1__TightPFIso__TightID__Central : weight__mu1__LoosePFIso__MediumID__Central) * ((mu1__pt < 30) ? weight__mu1__TightID__Trk__Central : weight__mu1__MediumID__Trk__Central) * ((mu2__pt < 30) ? weight__mu2__TightPFIso__TightID__Central : weight__mu2__LoosePFIso__MediumID__Central) * ((mu2__pt < 30) ? weight__mu2__TightID__Trk__Central : weight__mu2__MediumID__Trk__Central) * weight__TrgSF__singleMu__IsoMu24Central
Run3__2026	weight__gen * weight__lumi * weight__xs * weight__pu * weight__mu1__LoosePFIso__MediumID__Central * weight__mu1__MediumID__Trk__Central * weight__mu2__LoosePFIso__MediumID__Central * weight__mu2__MediumID__Trk__Central * weight__TrgSF__singleMu__IsoMu24Central